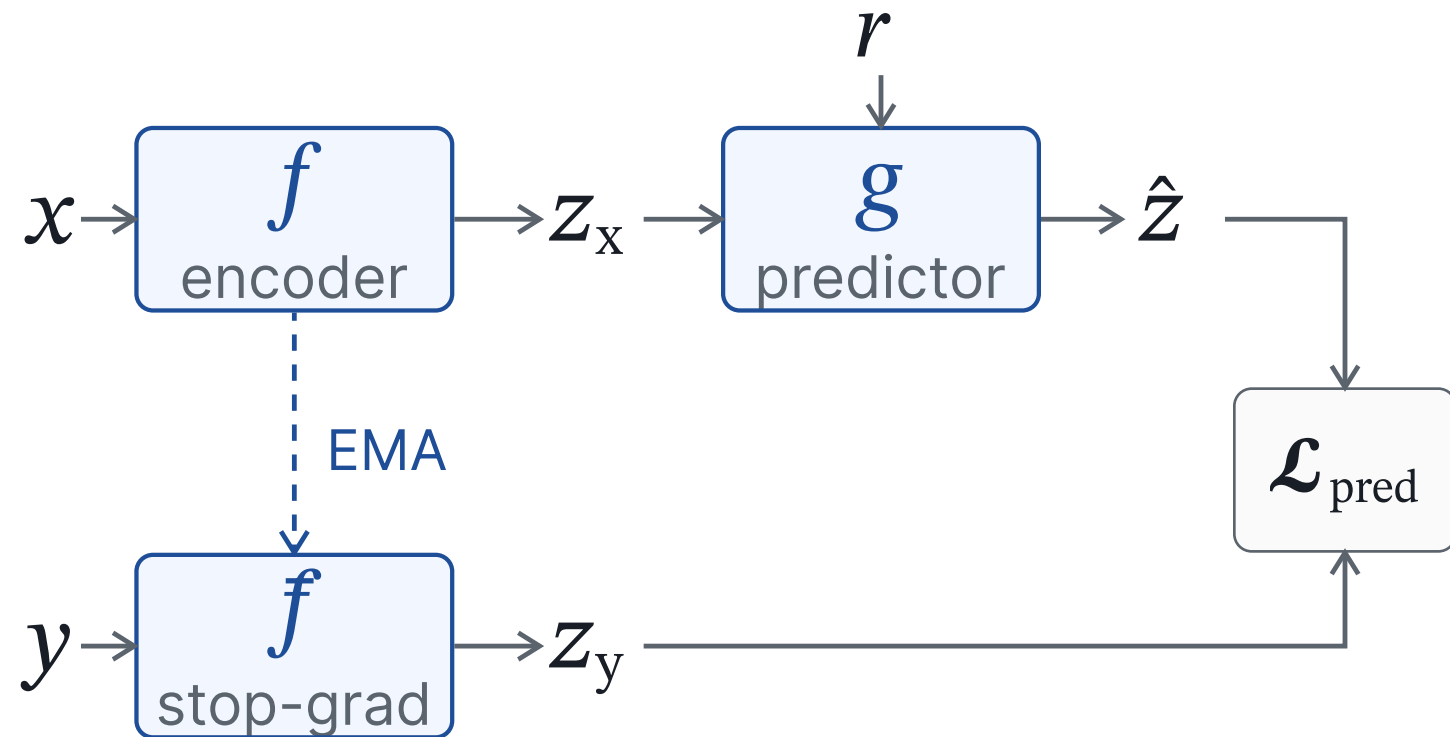


(a) AdaSSL reference

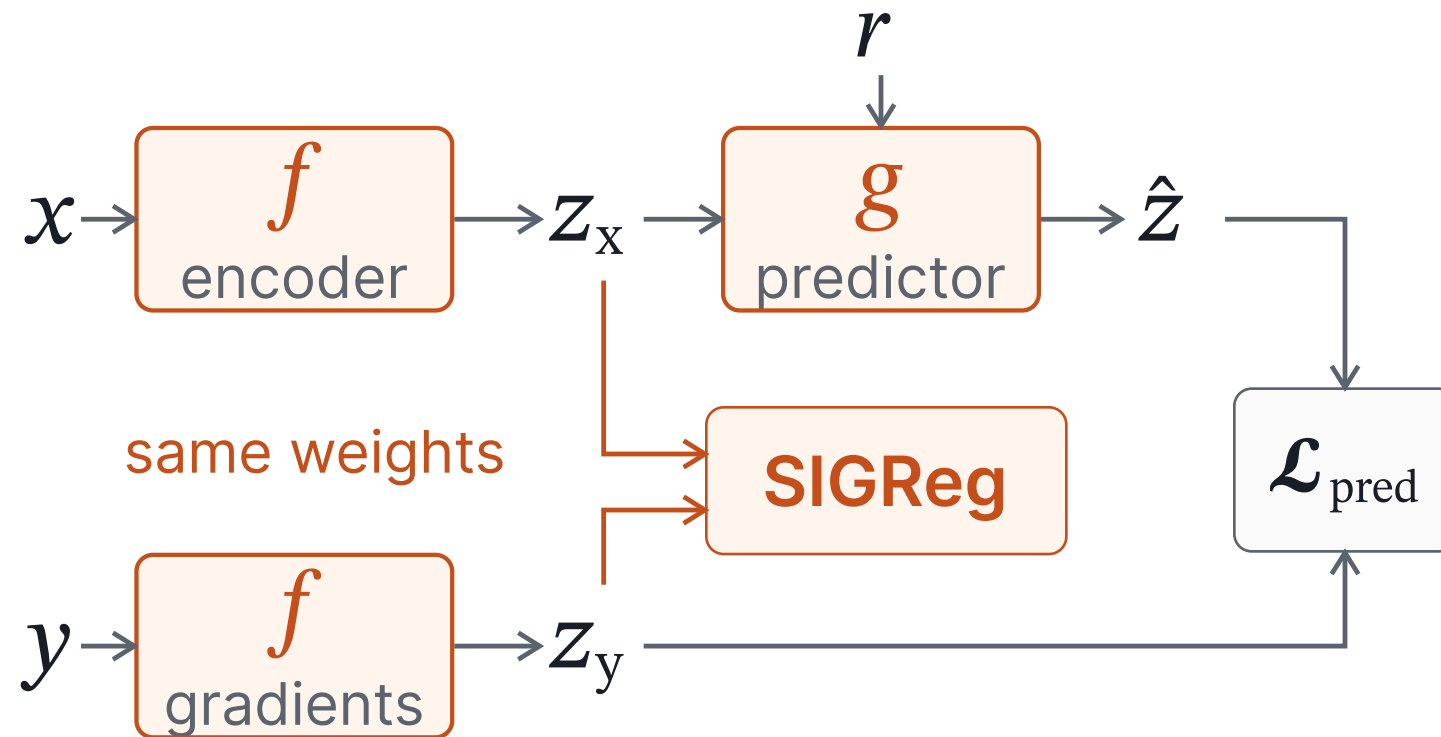
BYOL-style EMA target



$$\mathcal{L}_{\text{pred}} + \beta \mathcal{L}_{\text{KL}}$$

(b) Shared-Var (ours)

Shared encoder + SIGReg



$$\mathcal{L}_{\text{pred}} + \beta \mathcal{L}_{\text{KL}} + \lambda \mathcal{L}_{\text{SIG}}$$

Gaussian residual: $r \sim q_{\varphi}$ in training; $r \sim p_{\theta}$ at forecasting.